



NeuroScan - Brain Tumor Detection

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Python libraries and Tools

- kagglehub, PIL, os
 - Data loading, file organization
- Numpy
 - Data organization
- matplotlib, seaborn
 - Visualization of data
- sklearn.metrics
 - Calculation for accuracy, precision, etc.
- tensorflow.keras
 - Machine learning



What is NeuroScan?

- NeuroScan is a machine learning model built using *Convolutional Neural Networking*
- Its goal is to detect brain tumors in brain MRI scans
- A highly accurate model such as NeuroScan can help doctors by improving early detection of a tumor



Resource of Data

[Brain Tumor MRI Dataset | Kaggle](#)

The dataset used for this model was obtained from Kaggle



Intuition

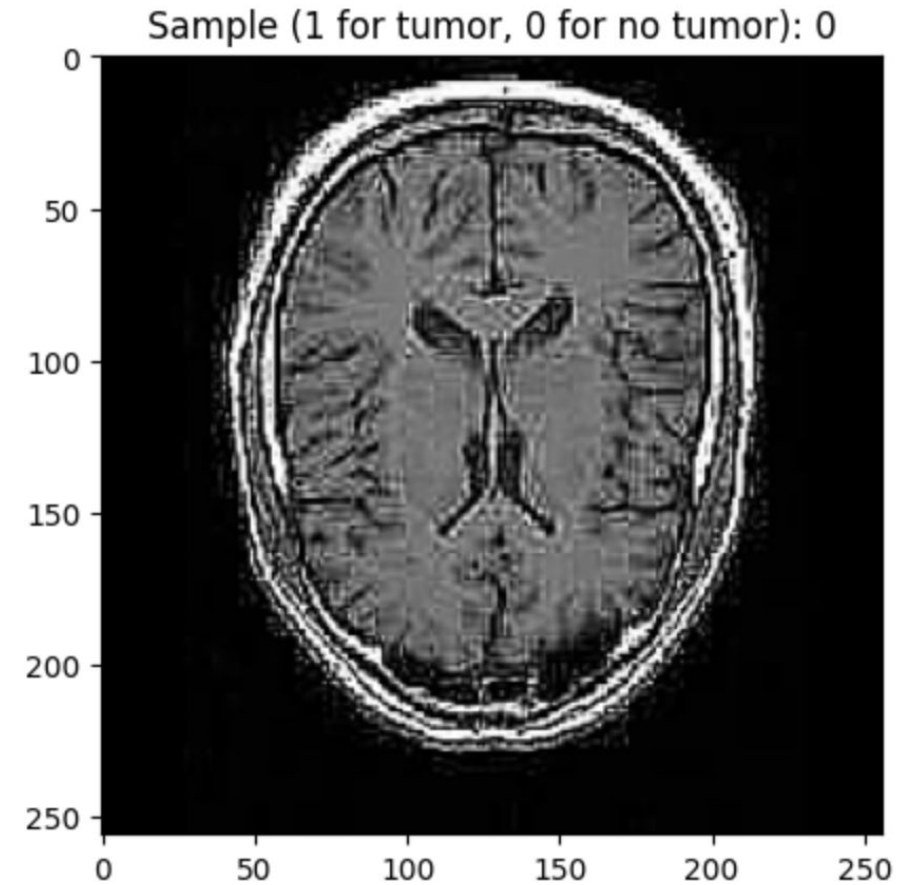
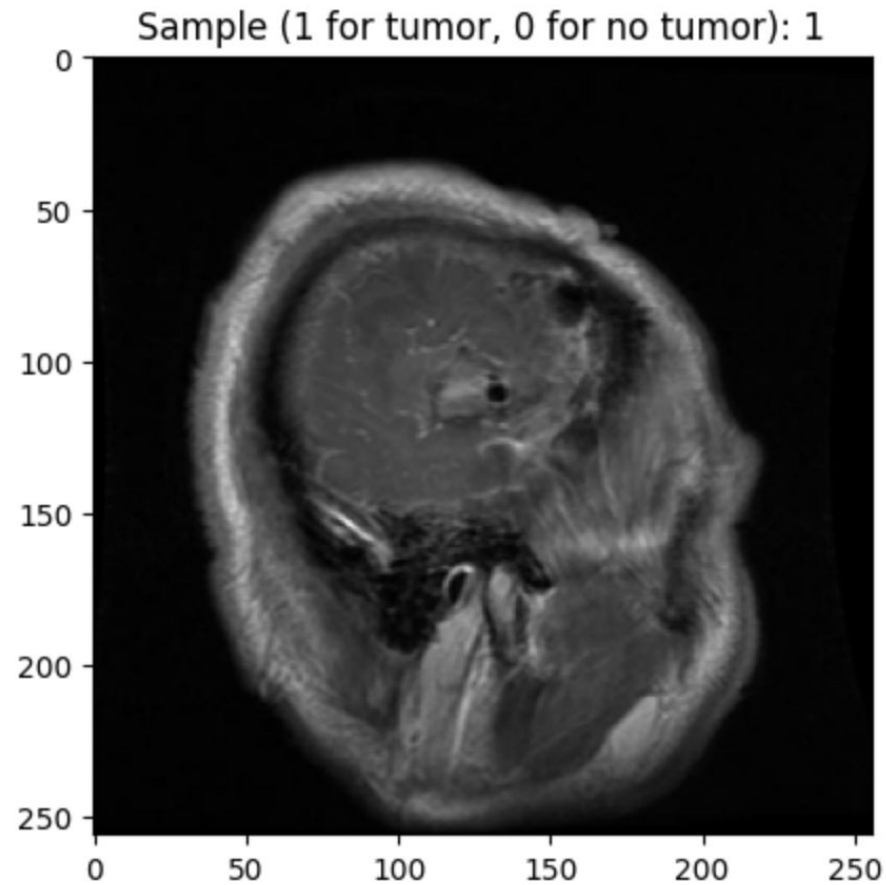
- Dataset used contains images of MRI scans
- Each MRI scan either has a tumor or no tumor at all (binary system)
- Although Kaggle Dataset has listed different types of tumor, we are simply just detecting if there is a tumor or not.



Data Preprocessing, cleaning, merging

- Loaded images from each class folder using PIL library
We need to access each image and open it correctly
- Converted all images to grayscale
Makes our model less complex, as MRI scans don't use RGB
- Resize every image to 256 x 256
We need to make sure every image is the same size so that the model can accept it
- converted each image into a Numpy Array
Turns the image into numerical data, allowing the model to work with it
- Put all converted image arrays into a dataset
combines the individual images into one dataset for our model to work with the data
- Gave each image a numeric label, 1 = tumor and 0 = no tumor
Did this to determine between tumor and no tumor images.

Data



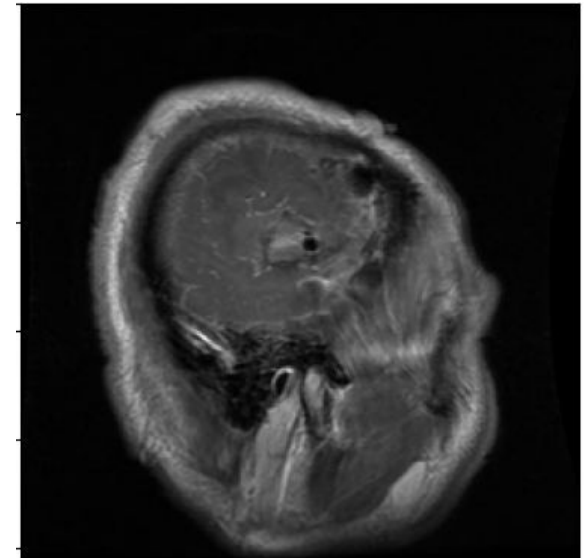


Desired Outcome

- Train a model that can detect a brain tumor based on the given MRI image.
- The accuracy need to be over 99.5% in order to use it in a medical situation.
- The false negatives (people with tumor diagnosed as non-tumor) should be near zero in order to not let people who actually have tumor to be diagnosed as non-tumor.

Choice of ML Model

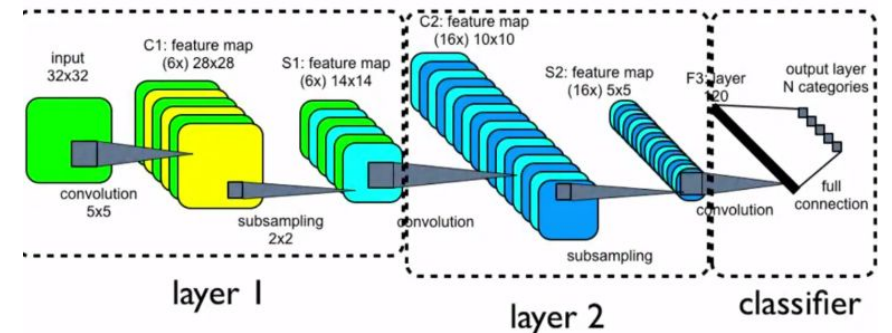
- We need binary classification (tumor vs non-tumor) of 2D images
- Out of CNN and KNN, CNN is a better choice because MRI images that we are training with comes in different orientations.
- Detection based on distance would be near impossible.
- CNN allows detection of brain tumor even when the image is flipped or have been taken in different orientations.



Training Parameters

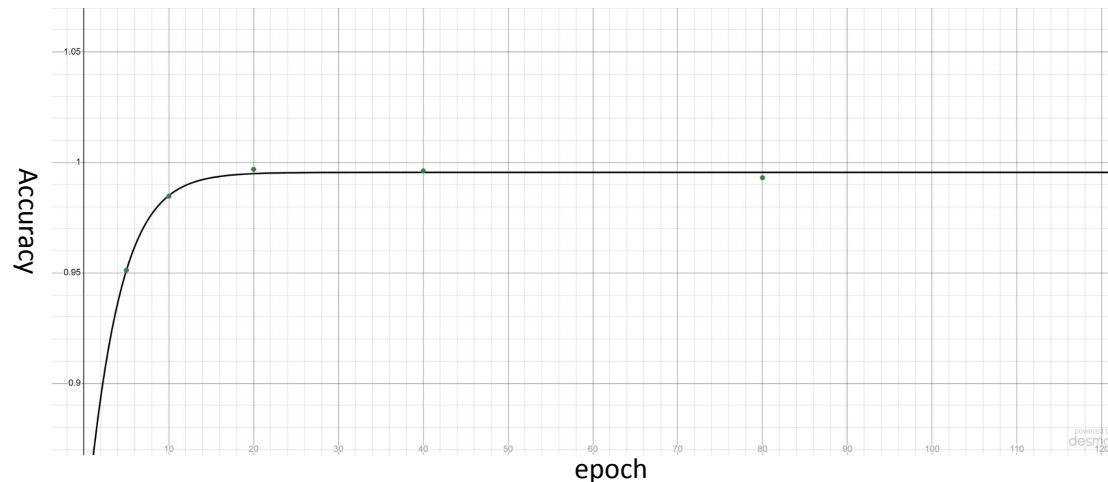
- Activation function: ReLU, Sigmoid
 - good for outputting probabilities and binary classification
- Learning Rate: 0.001
 - we wanted low and efficient rate
- Layers and Filters: 3 layers with each having 16, 32, 64 (3x3) filters
 - Rather than using complex designs like AlexNet, we wanted to use a simpler architecture.

Convolutional Neural Networks



Model Performance Overview

- 20 or more epochs resulted in an accuracy within .003 of .995
- Successfully identifies both tumor and non-tumor images with very few errors
- Demonstrates effectiveness of CNNs for medical image classification





Metrics (20 epochs)

Accuracy: .993 - Model is making very few incorrect predictions

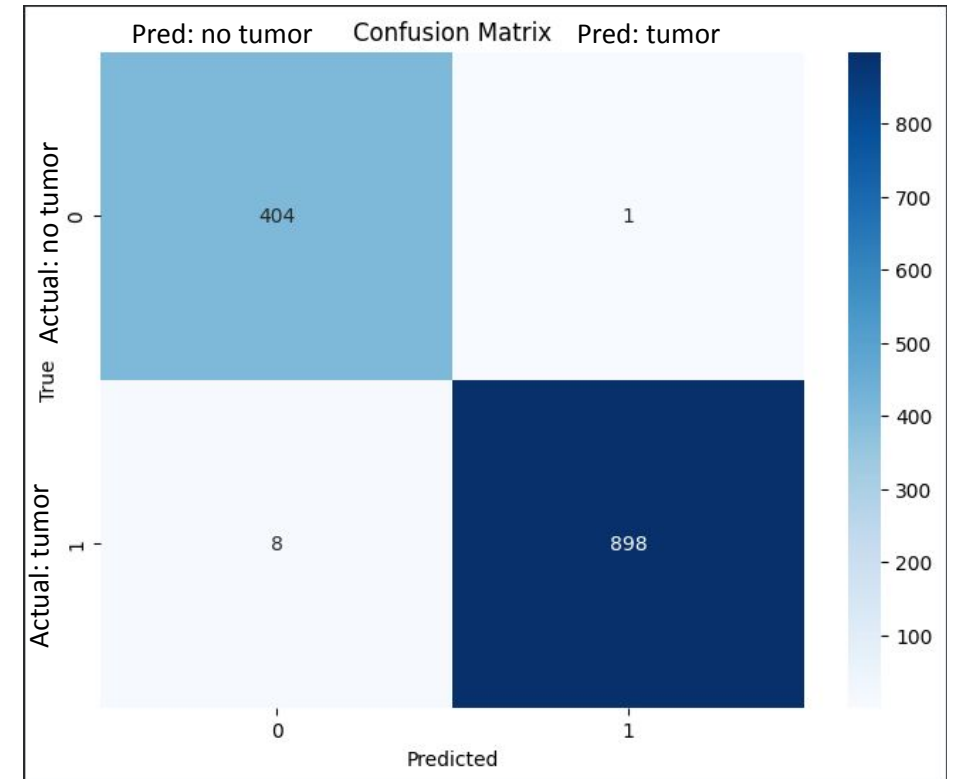
Precision: .994 - Model rarely miss-predicts a tumor when a tumor does not exist

Recall: .989 - Almost all cases where a tumor exist are flagged

F1: .992 - Good at catching tumors and good at avoiding false-positives (mixture of precision and recall)

Confusion Matrix

- 1 false-positive (predicted tumor but there is no tumor)
- 8 false-negatives (predicted no tumor but there was)
- 1302 images were predicted correctly





Conclusion

- Loaded MRI brain scan images
- Cleaned the data and processed it
- Assigned labels to the training data (1 = tumor, 0 = no tumor)
- Trained a model to determine whether a image contains a brain tumor or not.



Next Steps and Limitations

- Currently our model is limited to detecting whether or not the image contains a brain tumor (binary classification)
- In terms of a next step, we plan on implementing a multi-class classification system allowing it to identify which specific type of tumor the image contains
- We would like to improve our model to make sure that there are less false-negatives